

Project Name : Gold Trading Automation Platform (GTAP)

Document Name : Software Requirement Specification

Version: 1.0

Prepared By: Vinit Kushwaha

Date: 05 July 2026

Status: Draft

Software Requirement Specification (SRS)

Chapter 1 – Cover Page

Software Requirement Specification (SRS)

Project Name

Gold Trading Automation Platform (GTAP)

Document Type

Software Requirement Specification

Version

Version 1.0

Project Type

Web-Based Automated Gold Trading Platform

Prepared By

Vinit Kushwaha

Client

Confidential

Technology Stack

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js, Express.js
- Database: MongoDB
- Trading Engine: Python
- Trading Platform: MetaTrader 5 (MT5)
- Broker: XM
- Real-time Communication: Socket.IO
- Authentication: JWT
- Notifications: Telegram Bot API

Date

05 July 2026

 Chapter 2 – Document Control

2.1 Document Information

Field	Value
Document Name	Software Requirement Specification
Version	1.0
Status	Draft
Prepared By	Vinit Kushwaha
Project	Gold Trading Automation Platform

2.2 Version History

Version	Date	Description
1.0	05 July 2026	Initial Draft

2.3 Approval

Role	Status
Client	Pending
Developer	Pending

Chapter 3 – Introduction

3.1 Purpose

The purpose of this Software Requirement Specification (SRS) is to define the technical requirements, software architecture, system behavior, and implementation details for the Gold Trading Automation Platform.

This document serves as the primary technical reference for software development, testing, deployment, and future maintenance.

3.2 Project Scope

The software will automate Gold (XAUUSD) trading using a predefined scalping strategy on MetaTrader 5.

The system will:

- Monitor market data in real time.
 - Execute Buy/Sell orders automatically.
 - Manage Stop Loss and Take Profit.
 - Display real-time trading information through a web dashboard.
 - Send Telegram notifications.
 - Store all trading data securely in MongoDB.
-

3.3 Intended Audience

This document is intended for:

- Software Developers
 - Backend Developers
 - Python Developers
 - Test Engineers
 - System Administrator
 - Project Owner
-

3.4 Definitions

Term	Meaning
MT5	MetaTrader 5 Trading Platform
EMA	Exponential Moving Average
RSI	Relative Strength Index
Fractal	5-Candle Swing Detection
JWT	JSON Web Token
API	Application Programming Interface
VPS	Virtual Private Server
RR	Risk Reward Ratio

3.5 References

This software is based on:

- Client Trading Strategy
 - Business Requirement Specification (BRS v1.0)
 - MetaTrader 5 Integration
 - XM Broker Trading Rules
-



Chapter 4 – Overall Description

4.1 Product Perspective

The Gold Trading Automation Platform is a standalone web application integrated with MetaTrader 5.

The system consists of:

- Web Dashboard
 - Backend Server
 - MongoDB Database
 - Python Trading Engine
 - MT5 Terminal
 - Telegram Notification Service
-

4.2 Product Functions

The system shall provide:

- Secure Login
 - Dashboard
 - Trading Engine
 - Live Monitoring
 - Reports
 - Telegram Alerts
 - Risk Management
 - Strategy Execution
 - Trade History
 - System Logs
-

4.3 User Classes

Trader

Can:

- Login
 - View Dashboard
 - Start Trading Bot
 - Stop Trading Bot
 - View Reports
-

Future Admin

Can:

- Manage Users
 - Manage Brokers
 - View Logs
 - Monitor System
 - Configure Settings
-

4.4 Operating Environment

The application shall run on:

Server

- Ubuntu VPS
- Node.js
- Python
- MongoDB

Client

- Google Chrome
- Microsoft Edge
- Firefox

Trading Platform

- MetaTrader 5
-

4.5 Design Constraints

The system shall use:

- Node.js Backend
 - Express.js Framework
 - MongoDB Database
 - Python Trading Engine
 - MetaTrader 5
 - JWT Authentication
-

4.6 Assumptions

It is assumed that:

- MT5 is installed.
- User has an active XM account.
- VPS remains online 24×7.
- Internet connectivity is stable.
- Telegram Bot is properly configured.

Chapter 5 – System Architecture

5.1 Overview

The Gold Trading Automation Platform (GTAP) follows a modular, client-server architecture. The system is designed to separate the User Interface, Backend Services, Trading Engine, Database, and External Services to ensure scalability, maintainability, and reliability.

5.2 Architecture Components

Frontend

Technology: HTML, CSS, JavaScript

Responsibilities:

- User Login
 - Dashboard
 - Live Trade Monitoring
 - Reports
 - Settings
 - Bot Control (Start/Stop)
-

Backend Server

Technology: Node.js + Express.js

Responsibilities:

- User Authentication (JWT)
- API Management

- Communication with MongoDB
 - Communication with Python Trading Engine
 - Dashboard Data Processing
 - Logging
-

Database

Technology: MongoDB

Responsibilities:

- User Information
 - Trade History
 - Trading Logs
 - Bot Settings
 - Reports
 - Configuration
-

Trading Engine

Technology: Python

Responsibilities:

- MT5 Connection
 - Market Data Monitoring
 - EMA Calculation
 - RSI Calculation
 - Fractal Detection
 - Buy/Sell Execution
 - Stop Loss Management
 - Take Profit Management
 - Trailing Stop Loss
-

MetaTrader 5

Responsibilities:

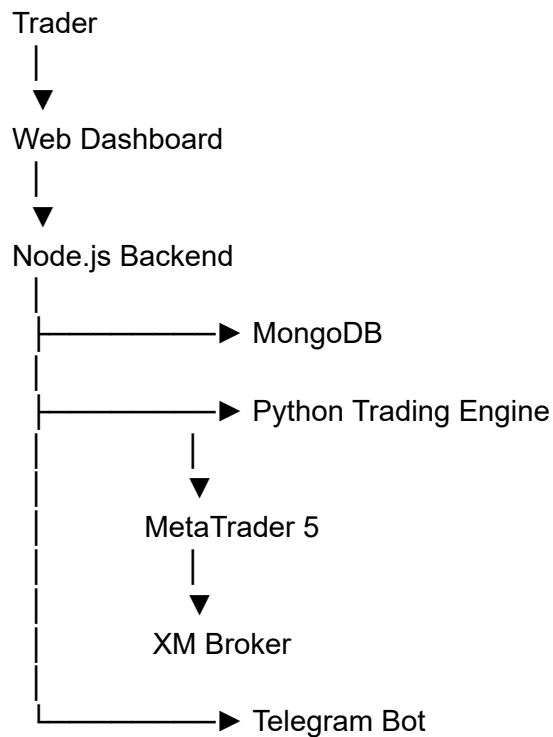
- Execute Orders
 - Receive Market Data
 - Maintain Trading Account
-

Telegram Bot

Responsibilities:

- Trade Alerts
 - Error Alerts
 - Bot Status
 - Daily Summary
-

5.3 System Architecture Flow



Chapter 6 – Functional Modules

The system is divided into the following modules.

Module 1 – Authentication

Functions:

- Login
- Logout

- JWT Authentication
 - Session Validation
-

Module 2 – Dashboard

Functions:

- Balance
 - Equity
 - Margin
 - Active Trades
 - Closed Trades
 - Win Rate
 - Drawdown
 - Daily Profit
 - Monthly Profit
-

Module 3 – Trading Engine

Functions:

- Market Monitoring
 - Strategy Execution
 - Buy Order
 - Sell Order
 - Stop Loss
 - Take Profit
 - Trailing Stop Loss
-

Module 4 – Reports

Functions:

- Daily Report
 - Weekly Report
 - Monthly Report
 - Trade History
 - Performance Report
-

Module 5 – Notifications

Functions:

- Telegram Alerts
 - Error Notifications
 - Trade Notifications
 - Bot Status
-

Module 6 – Settings

Functions:

- MT5 Credentials
 - Trading Parameters
 - Risk Settings
 - Telegram Configuration
-

Chapter 7 – Database Design (High-Level)

The system uses MongoDB.

Collection 1 – Users

Fields:

- User ID
 - Name
 - Email
 - Password
 - Role
 - Status
-

Collection 2 – Trades

Fields:

- Trade ID

- Symbol
 - Order Type
 - Entry Price
 - Exit Price
 - Stop Loss
 - Take Profit
 - Lot Size
 - Profit/Loss
 - Status
 - Open Time
 - Close Time
-

Collection 3 – Bot Settings

Fields:

- EMA
 - RSI
 - Lot Size
 - Risk Reward
 - Max Trades
 - Trailing SL
 - Telegram Status
-

Collection 4 – Logs

Fields:

- Log ID
 - Log Type
 - Message
 - Time
-

Collection 5 – Reports

Fields:

- Daily Profit
- Weekly Profit
- Monthly Profit
- Win Rate
- Drawdown



Chapter 8 – API Specification

The backend exposes REST APIs.

Authentication APIs

POST /api/auth/login

POST /api/auth/logout

Dashboard APIs

GET /api/dashboard

GET /api/dashboard/statistics

GET /api/dashboard/trades

Trading APIs

POST /api/trading/start

POST /api/trading/stop

GET /api/trading/status

GET /api/trading/history

Settings APIs

GET /api/settings

PUT /api/settings

Reports APIs

GET /api/reports/daily

GET /api/reports/weekly

GET /api/reports/monthly

Bot APIs

POST /api/bot/start

POST /api/bot/stop

GET /api/bot/status

Chapter 9 – External Interface Requirements

यह Chapter बताता है कि हमारा Software किन External Systems से जुड़ेगा।

9.1 MetaTrader 5 (MT5) Interface

Purpose

Python Trading Engine MT5 Terminal से Connect होकर Market Data पढ़ेगा और Trades Execute करेगा।

Data Exchange

Input:

- Live Market Price
- Candle Data
- Account Information
- Open Positions

Output:

- Buy Order
 - Sell Order
 - Stop Loss Update
 - Take Profit Update
 - Trade Close
-

9.2 MongoDB Interface

MongoDB निम्न Data Store करेगा:

- User Details
- Trade History
- Reports
- Bot Settings
- Logs
- Performance Statistics

9.3 Telegram Bot API

Telegram Bot निम्न Notifications भेजेगा:

- Bot Started
 - Bot Stopped
 - Buy Executed
 - Sell Executed
 - Stop Loss Hit
 - Take Profit Hit
 - MT5 Connection Lost
 - Connection Restored
 - Daily Summary
-

9.4 Frontend Interface

Frontend और Backend REST APIs तथा Socket.IO के माध्यम से Communication करेंगे।

Functions

- Login
 - Dashboard
 - Reports
 - Settings
 - Live Trade Updates
-

9.5 Broker Interface

Current Broker:

- XM Broker

Future Support:

- Exness
 - IC Markets
 - Pepperstone
-



Chapter 10 – Security Requirements

Security इस Project का सबसे महत्वपूर्ण भाग है।

Authentication

- JWT Authentication
 - Secure Login
 - Session Validation
 - Token Expiration
-

Authorization

Role Based Access

Current Roles:

- Trader

Future Roles:

- Admin
 - Support
-

Password Security

- Password Hashing
 - Strong Password Policy
-

API Security

- JWT Protected APIs
 - Input Validation
 - Error Handling
 - Rate Limiting (Future)
-

Database Security

- Restricted Database Access
 - Environment Variables
 - Secure Connection
-

Logging

System Logs

Trade Logs

Error Logs

Audit Logs

Chapter 11 – Performance Requirements

The platform should deliver reliable and fast performance.

Trading Performance

- Fast Order Execution
 - Low Latency
 - Real-Time Monitoring
-

Dashboard Performance

- Live Updates
 - Responsive UI
 - Instant Data Refresh
-

Backend Performance

- Efficient API Response
 - Optimized Database Queries
 - Proper Error Handling
-

System Availability

- 24×7 Operation
 - Automatic Recovery
 - Auto Reconnection
-

Scalability

Future Support

- Multiple Users
 - Multiple Accounts
 - Multiple Brokers
 - AI Strategy Engine
-



Chapter 12 – Future Enhancements

The architecture should support future upgrades without major redesign.

Planned Features

User Management

- Multiple Users
 - User Roles
 - Permissions
-

Trading

- Multiple Strategies
 - Strategy Switching
 - Strategy Marketplace
-

Broker Support

- Exness
 - IC Markets
 - Pepperstone
 - Binance (Future)
-

Dashboard

- Advanced Charts
 - Performance Analytics
 - Portfolio Analysis
 - Drawdown Graphs
-

Notifications

- Email Notifications
 - WhatsApp Notifications
 - Push Notifications
-

Mobile

- Android Application
 - iOS Application
-

Artificial Intelligence

- AI Trade Analysis
 - AI Trade Suggestions
 - Performance Prediction
-

Commercial Features

- Subscription Plans
- Payment Gateway
- License Management